D-Modem

AT Command Reference

The modem core of slmodemd, libdmodem and the PBX modules

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1 Introduction

D-Modem's modem core (`core/` in the source) takes the AT commands of every D-Modem program:

- **slmodemd**, the modem on a pseudo terminal (`/dev/ttyDM0` links to it), its line a SIP call;
- **libdmodem** (`dmodem.h`), the modem as a library: the program that embeds it sends the commands;
- the **Asterisk and FreeSWITCH modules** (`app_dmodem`, `mod_dmodem`), whose profiles give the commands each call starts with.

The command set is D-Modem's own. It began as Smart Link's slmodemd's (the reference of Smart Link's ST56 modems is kept as `docs/legacy/ST56ATCommands.pdf`) and has grown V.90 and V.92 as the digital modem, V.91, V.44 and V.92's modem-on-hold since. It takes as well the commands that other modems' setup strings send (V.250's, Rockwell's and Conexant's, USRobotics'), most of them with no effect, so that a setup string written for another modem works: section 10 lists them, and where D-Modem's meaning of a command differs from another modem's. Section 9 tells what a SIP line changes in slmodemd.

This reference follows the core as it is: every command, parameter range, default and output here is the core's own. A setting that is kept but changes nothing says so.

Notation. `AT` commands are shown in capitals; the modem takes either case. *n* is a decimal number, *h* a hexadecimal one; `[ ]` marks what may be left out. A setting's default is the one `ATZ` and the start of the program give.

2 Command lines

- A command line begins with `AT` and ends with the character of S3 (carriage return) or S4 (line feed). Blanks and tabs in it are dropped before it is read.
- The character of S5 (backspace) takes back the last one typed.
- A line holds 63 characters at most; a longer one is ERROR.
- With echo on (`E1`), what the DTE types comes back as it is typed.
- A line that does not begin with `AT` is not a command: with echo on it is blanked out on the DTE's screen.
- `A/` (no line end needed) runs the last command line again.

The commands of a line run in turn, one after the other (`ATE1Q0V1`); a basic command's number left out is 0 (`ATE` is `ATE0`). An extended command (`+...`) takes =*values*, `?` (its values) and `=?` (its ranges). A `;` may stand between two commands, as V.250 puts one after an extended command that another follows (`AT+IFC=2,2;+MS=V92`); here it is not needed. The first command that fails ends the line with ERROR; otherwise OK follows the last one. `A`, `D` and `O` end the line and give their result later (CONNECT, NO CARRIER and so on), and `Z` ends it with OK: whatever follows them on the line is not read. A command that reports something (`I`, `S`*n*`?`, `+MS?` ...) prints it on lines of its own before the result.

A character from the DTE while a call is being set up (dialing, answering, the modems' training) hangs up.

3 Result codes

V0	V1	
0	OK	the command line ran
1	CONNECT [*rate*]	the call is up, or a fax command's data begins
2	RING	a call rings (slmodemd: every 3 s, section 9)
3	NO CARRIER	no call came up, or the call is over
4	ERROR	a command failed, or a line too long
6	NO DIALTONE	no dial tone (X2, X4)
7	BUSY	a busy tone
8	NO ANSWER	no answer within S7 seconds of dialing

Q1 turns them off, V0 gives the numbers, each on a line of its own. With V1 a data call's CONNECT gives the rate the modem receives at, in bit/s (CONNECT 33600); with X0, in fax and in voice mode it is CONNECT alone.

Before CONNECT, &A (S70) adds lines on the connection (verbose results only), each bit one of them:

bit	line	
0 (1)	Modulation: *name*	the modulation's name (V92, V34, V32bis, ...)
1 (2)	Protocol: LAPM or NONE	V.42 error correction
0 and 1 (3)	Compression: V44, V42bis or NONE	what V.42 agreed
2 (4)	TxRate: *rate*	the rate the modem sends at

4 Basic commands

ATA

Answers: off hook, and the answering modem's tones after S92 seconds. In slmodemd the call that rings (RING) is answered. Its result is CONNECT, or NO CARRIER.

ATB, ATB*n*

ATB alone: V.22 at 1200 bit/s only, without automode, AT+MS=22,0,1200,1200. B0 to B3, B15 and B16, other modems' choice between the ITU and the Bell modulation at 1200 and 300 bit/s: taken, changing nothing (the call's negotiation and +MS choose).

ATD*string*

Dials *string* and calls. The dialer sends the digits and * # A B C D as DTMF (S11 ms each, as long between them); a comma pauses S8 seconds, W two; other characters are skipped. T and P choose tone or pulse dialing (S16) for the rest of the call. ATDL dials the last number again (Dialing *number*... first). With X2 or X4 the modem waits for a dial tone first. The result is CONNECT, BUSY, NO ANSWER, NO DIALTONE or NO CARRIER. In slmodemd the dial string is the SIP call's number (section 9).

ATE[*n*]

Echo of the DTE's characters in command mode: E0 off, E1 on (S13; default 1).

ATH[0], ATH1

H or H0 hangs up (on hook); H1 goes off hook.

ATI[*n*]

Information:

I0	the product: D-Modem
I1, I2	the product and version (D-Modem, 1.0.0), then the maker, D-Modem project
I3	the device's name, then the driver's (slmodemd: /dev/slamr0, socket driver)
I4	S0 to S23, eight a line (s00=000 s01=000 ...)
I5, I6	Stored Profile 0:, 1: (there are none)
I7	the country: Country: USA

Any other number is ERROR.

ATL[*n*], ATM[*n*]

The speaker's volume, L0 to L3 (S30; default 3), and when it is on, M0 never, M1 until the call is up, M2 always, M3 as M1 (on other modems not while dialing) (S29; default 1). A SIP line has no speaker: kept for the commands that set them.

ATN[*n*]

Automode (S31; default 1): N1 lets the modems find their best common modulation from the one +MS chose down, N0 keeps to it. The same as +MS's second parameter.

ATO

Back online after the escape (section 8): CONNECT. ERROR when there is no call to go back to.

ATP, ATT

Pulse or tone dialing (S16; default tone).

ATQ[*n*]

Q0 gives the result codes, Q1 keeps them back (S14; default 0).

ATS*n*?, ATS*n*=*v*

Reads S-register *n* (0 to 255), in decimal on a line of its own, or sets it to *v* (8 bits: *v* modulo 256). A number over 255 is ERROR. Section 7 lists them.

ATV[*n*]

V0 gives the result codes as numbers, V1 as words (S15; default 1).

ATX[*n*]

Results and dial tone, X0 to X4 (S56; default 4):

	CONNECT	before dialing
X0	without the rate	no wait for a dial tone
X1, X3	with the rate	no wait for a dial tone
X2, X4	with the rate	a dial tone of 1 s, NO DIALTONE if none in S6 seconds (2 s at least)

BUSY (four cadences of 0.2 to 0.8 s) and NO ANSWER come in all of them. A SIP call has no dial tone: slmodemd wants ATX3.

ATY[*n*]

Y0 to Y3: kept in S161, nothing more.

ATZ[*n*]

Z or Z0 to Z3: hangs up, and the S-registers, +MS, modem-on-hold, the data mode (+FCLASS=0) and the parameters kept for other modems' setup strings (section 10) as at the start; the country stays. There are no stored profiles: the number changes nothing. OK, and the rest of the line is not read.

5 Ampersand, backslash and percent commands

AT&A[*n*]

&A0 to &A7: the lines before CONNECT (S70; default 0), see the result codes.

AT&C[*n*], AT&E[*n*], AT&H[*n*]

Data carrier detect (&C0, &C1; S60), the rate CONNECT gives (&E0, &E1; S71) and flow control (&H0 to &H3; S62): kept, but they change nothing (a pty has no carrier line, CONNECT always gives the receive rate, and flow control is the pty's). Default 0.

AT&D[*n*], AT&F[*n*], AT&R[*n*], AT&S[*n*]

Taken, with or without a digit, and nothing done (ATZ gives the defaults).

AT&K[*n*], AT%C[*n*]

Data compression (S104; default 3): 0 none, 1 to 3 V.44 or V.42bis both ways, as the other modem agrees (V.42 offers both). It needs V.42 (\N). &K is compression as on USRobotics' modems; Rockwell's and Conexant's use it for flow control, whose &K4 to &K6 are taken with no effect, and whose &K3 (RTS/CTS) leaves the compression on.

AT&P[*n*]

&P0 to &P3: the pulse dialing ratio, kept in S28.

AT&Q[*n*]

Other modems' choice of error correction, as \N: &Q0 and &Q6 are \N0 (none), &Q5, &Q8 and &Q9 are \N3 (V.42 when the other modem has it; there is no MNP). The others (synchronous modes) are ERROR.

AT&V, AT&W[*n*]

There are no stored profiles: &V (show them) is ERROR, and &W, &W0, &W1 (store one) answer OK, storing nothing.

AT\A[*n*]

\A0 to \A3, MNP's block size: taken (there is no MNP); \A4 and up are ERROR.

AT\N[*n*]

Error correction (S103; default 1): \N0 none; \N1 to \N5 V.42 (LAPM) when the other modem has it, a call without when it does not (V.42's detection phase decides). There is no MNP: an MNP modem that has V.42 connects with LAPM, one with MNP only without error correction or not at all.

AT%E[*n*]

%E0 to %E2, line quality monitoring: kept in S39 (the modulations retrain and renegotiate their rate by themselves).

AT*B, AT%B

The numbers barred from dialing: none are kept; OK.

AT#CID=*n*, AT#CID?, AT#CID=?, AT+VCID...

Caller ID between the first two rings: 0 off, 1 or 2 on (=? gives 0,1,2). +VCID (V.253's) is the same, in every mode. A SIP call rings without it.

6 Extended commands

6.1 Modulation: +MS

`AT+MS=[`*mod*`][,[`*automode*`][,[`*min*`][,[`*max*`]]]]`

The modulation the modem calls or answers with, whether it may fall back (*automode* 1, as `N1`; `S31`) and the rates it may connect at, *min* to *max* bit/s (300 to 56000, *min* no more than *max*). *mod* is a number or V.250's name for it, in either case (`AT+MS=V34` is `AT+MS=34`); `?` and `=?` give the numbers. A parameter left out keeps its value: `AT+MS=,0` only turns automode off. The default is `92,1,300,56000`.

mod	**name**		
21	`V21`	ITU V.21	300 bit/s FSK
23	`V23C`	ITU V.23	1200/75 bit/s FSK
103	`B103`, `Bell103`	Bell 103	300 bit/s FSK
22	`V22`	ITU V.22	1200 bit/s DPSK
212	`B212`, `Bell212A`	Bell 212A	1200 bit/s DPSK
122	`V22B`	ITU V.22bis	2400 bit/s, V.22 as its fallback
32	`V32`	ITU V.32	4800 and 9600 bit/s
132	`V32B`	ITU V.32bis	4800 to 14400 bit/s
34	`V34`	ITU V.34	2400 to 33600 bit/s, after V.8
90	`V90`	ITU V.90	as the digital modem, after V.8
91	`V91`	ITU V.91	two digital modems, after V.8
92	`V92`	ITU V.92	V.90 with quick connect, PCM upstream and modem-on-hold
56	`K56`	K56flex	not implemented: runs V.90

V.90 and V.92 are there as the *digital* modem (the server's side, as on a G.711 VoIP line), on an 8000 Hz G.711 line: analogue V.90 and V.92 modems calling connect at up to 56000 bit/s downstream. 90 and 92 offer V.91 too, which two digital modems use instead, and V.34 for the others; on a line at another sample rate they are V.34. With automode V.8 and the V.34 modem hand over to V.32bis (and its V.22bis and V.22) for a modem without V.34, and the answering modem tries V.23 and Bell 103 for one without V.8.

`AT+MS?`

The settings: `92,1,300,56000`.

`AT+MS=?`

What may be chosen:
`(21,22,23,122,32,132,34,103,212,56,90,91,92),(0,1),(300-56000),(300-56000)`

6.2 Identification and country

`AT+GMI, AT+GMM, AT+GMR`

The manufacturer (`D-Modem project`), the model (`D-Modem`) and the revision (`1.0.0`, the build's version); with `?` after them (`AT+GMI?`) the same.

`AT+GCAP`

The command sets the modem has: `+GCAP: +FCLASS,+MS`.

`AT+GCI=`*h*`, AT+GCI?, AT+GCI=?`

The country by its T.35 code in hexadecimal (`+GCI:b5`: USA), which sets its dial pause (S8) and hook flash (S24) times; `=?` lists the codes. slmodemd's `-c` option names the country it starts with (`-countrylist` lists them).

6.3 Mode

`AT+FCLASS=`*n*`, AT+FCLASS?, AT+FCLASS=?`

`0` data (default), `1` fax class 1, `8` voice (`=?` gives `0,1,8`).

6.4 V.92 modem-on-hold

`AT+PMH=`*n*`, AT+PMH?, AT+PMH=?`

`0` lets the other modem put the call on hold, `1` refuses (default 0; `+PMH?` gives `+PMH: 0`).

`AT+PMHT=`*n*`, AT+PMHT?, AT+PMHT=?`

How long a hold the modem grants: the code of V.92's Table 33, `0` to `13` (`0` grants none; default 5).

`AT+PMHR[=`*s*`]`

In online command mode on a V.92 connection: asks the other modem to go on hold, for *s* seconds (1 to 960; 10 if left out). ERROR otherwise.

6.5 Fax class 1

With `AT+FCLASS=1`, the modem takes T.31's class 1 commands, the fax machine's side being the DTE's (the PBX modules and libdmodem have a T.30 engine of their own as well). `+FAA` and `+FPR` are kept with no effect (section 10).

`AT+FMI?, AT+FMM?, AT+FMR?`

The manufacturer, the model and the revision, as `+GMI`, `+GMM` and `+GMR` give them (in any mode).

`AT+FTS=`*n*`, AT+FRS=`*n*

Sends silence for, or waits for silence of, *n* times 10 ms.

`AT+FTM=`*m*`, AT+FRM=`*m*`, AT+FTM=?, AT+FRM=?`

Sends or receives data at modulation *m*: CONNECT, then the data with <DLE> doubled, ended by <DLE><ETX>.

m	
24, 48	V.27ter, 2400 and 4800 bit/s
72, 96	V.29, 7200 and 9600 bit/s
73, 97, 121, 145	V.17, 7200 to 14400 bit/s, long training
74, 98, 122, 146	V.17, 7200 to 14400 bit/s, short training

`AT+FTH=3, AT+FRH=3, AT+FTH=?, AT+FRH=?`

Sends or receives HDLC frames at V.21 channel 2 (300 bit/s), the only one: `=?` gives `3`.

6.6 Voice

With AT+FCLASS=8 the modem takes these voice commands (ERROR in the other modes):

command	
+VIP	the voice settings as at the start
+VNH=*n*	0 or 1
+VGT=*n* , +VGR=*n*	the gain sending and receiving, 0 to 128
+VIT=*n*	the inactivity timer, 0 to 255
+VSD=*s* , *t*	silence detection: sensitivity and period, 0 to 255
+VSM=*m* ,8000	the compression method, 0 or 1, at 8000 samples/s
+VLS=*n*	1 takes the line for voice, 0 leaves it
+VTX, +VRX, +VTR	sends, receives, or both
+VTS=[*f1* , *f2* , *d*]	a tone of *f1* and *f2* Hz for *d* (0 to 6000)
+VTS={*s* , *d*}, +VTS=*s*	the DTMF digit *s* (for *d*, 0 to 100)

Each takes ? (its values) and =? (their ranges). +VCID, caller ID, is #CID's, in every mode.

7 S-registers

Registers 0 to 255 hold 8 bits each. Those not listed are 0 and free; “kept” means set and read back, with no other effect.

S		default	
0	rings to answer after (0: never)	0	
1	rings so far	0	
2	the escape character	43 (+)	
3	the line’s end	13 (CR)	
4	the line feed	10 (LF)	
5	backspace	8 (BS)	
6	wait for a dial tone, s	2	X2, X4; 2 s at least
7	wait for the answer and the training, s	60	
8	a comma’s pause, s	2	the country’s
9	carrier detect time, 0.1 s	6	kept
10	carrier loss before hanging up, 0.1 s	7	
11	DTMF tone and gap, ms	100	below 40: 95
12	escape guard time, 20 ms	50	kept: the guard is 0.5 s
13	echo (E)	1	
14	quiet (Q)	0	
15	verbose (V)	1	
16	tone (1) or pulse (0) dialing (T, P)	1	
24	hook flash time, 10 ms	20	the country’s
28	pulse ratio (&P)	0	kept
29	speaker control (M)	1	
30	speaker volume (L)	3	
31	automode (N, +MS)	1	
32	the modulation (+MS)	92	
39	line quality (%E)	0	kept
56	results and dial tone (X)	4	
60	carrier detect (&C)	0	kept
62	flow control (&H)	0	kept
70	CONNECT’s lines (&A)	0	
71	CONNECT’s rate (&E)	0	kept
92	the answer’s delay, s (0: 10 ms)	2	before the answer tone
103	error correction (\N)	1	
104	data compression (%C, &K)	3	
161	Y	0	kept

8 Online, the escape and hanging up

Online, what the DTE sends goes to the other modem, except the escape: three S2 characters (+++), each within 0.5 s of the one before, with 0.5 s of nothing before them and after them. The modem then says OK and is in online command mode, the call still up: `ATO` goes back online, `ATH` hangs up, and on a V.92 connection `AT+PMHR` asks for modem-on-hold.

The call ends with NO CARRIER when the other side hangs up or its carrier is gone for S10 tenths of a second. slmodemd hangs up as well when the DTE drops DTR (it sets the pty's speed to 0) or, with HUPCL set on the pty, closes it; it then makes a new pty (`/dev/ttyDM0` follows it).

9 slmodemd on a SIP line

- `ATD`*number* calls `sip:`*number*`@`*domain* of `SIP_LOGIN`. A telephone number (only V.250's dial characters: digits, `* # + A B C D`, the modifiers `T P W , @ !` `;` and `- ( ) .` and blanks) goes as its digits and DTMF characters: `ATDT555-1000` calls `sip:5551000@`*domain*. Anything else is a SIP user's name, as it is: `ATDalice`.
- The line has no dial tone: `ATX3` (or `X1`), else NO DIALTONE.
- With `-L` slmodemd registers and takes calls: RING every 3 seconds while one waits, `ATA` or `S0` rings (`ATS0=1`) answer it, and the rings stop if the caller gives up. A call that comes while the line is in use is refused (486, busy); without `-L` every call is (480).
- The call is G.711 at 8000 Hz, which is what V.90 and V.92 need (`AT+MS=92`, the default). Over a real PSTN gateway a lower modulation (`AT+MS=132,0,4800,9600`) may do better.

10 Setup strings from other modems

A setup string written for another modem (a program's, a driver's, a guide's) holds commands of V.250, of Rockwell's and Conexant's modems, of USRobotics' and of Hayes'. The modem takes the common ones: those it has a meaning for do what the sections above say, and the others, below, are taken with no effect, so that the line ends in OK. Flow control is the pty's own and the line a G.711 call, and the modulations retrain and renegotiate by themselves: what these commands set elsewhere has no use here.

10.1 Basic commands taken

Each with the values listed (a number left out is 0), OK and nothing more; another value is ERROR.

command	on other modems
`B0 to B3, B15, B16`	the ITU or the Bell modulation at 1200 and 300 bit/s (`ATB` alone: V.22 here)
`C1`	the carrier on (`C0`, receive only, is ERROR)
`F1`	no echo of the data online (`F0` is ERROR)
`W0 to W2`	what CONNECT reports (here `&A` and `&E`)
`&B0 to &B2`	V.32's retrain; USRobotics': the DTE's rate fixed
`&G0 to &G2`	V.22bis's guard tone
`&I0 to &I5`	USRobotics' XON/XOFF flow control
`&J0, &J1`	the auxiliary relay
`&K4 to &K6`	Rockwell's flow control (`&K0 to &K3`: compression here)
`&M0, &M4, &M5`	asynchronous mode; USRobotics': error correction (here `\N`)
`&N0 to &N14`	USRobotics': the line's rate (here `+MS`)
`&W0, &W1`	store the profile (there are none)
`&Y0, &Y1`	the profile at power on
`\G0, \G1`	XON/XOFF passed to the other modem
`\J0, \J1`	the DTE's rate following the line's
`\K0 to \K5`	what a break does
`\Q0 to \Q3`	flow control
`\R0, \R1`	the ring indicator after the answer
`\T0 to \T255`	the inactivity timer
`\V0 to \V2`	the protocol in CONNECT (here `&A`)
`-C0, -C1`	the calling tone
`-K0 to -K2, -Q0, -Q1, )M0 to )M2, @M0 to @M30, *H0 to *H2`	Rockwell's MNP 10 and 10EC (cellular): MNP Extended Services, the V.22bis fallback, power levels, the link's speed

`&D, &F, &R, &S` and `\A` are taken as well (section 5).

10.2 Extended parameters kept

These keep their values with no other effect: =*values* sets them (each in its range, else ERROR; one left out stays as it is), ? gives +*name*: *values* (`-SEC?` the values alone), =? the ranges, and `ATZ` the defaults.

name	default	on other modems
`+IFC`	`2,2`	flow control, each way (0 to 3, 0 to 2)
`+ICF`	`3,3`	the DTE's character framing

name	default	on other modems
+IPR	0	the DTE's rate (0 automatic, 300 to 921600)
+ILRR	0	the DTE's rate reported
+IDSR	0	DSR
+ILSD	0	long space disconnect
+MR	0	the modulation reported (+MCR:, +MRR:)
+ER	0	the error correction reported (+ER:)
+DR	0	the compression reported (+DR:)
+ES	3,0,2	the error correction (here \N)
+DS	3,0,2048,32	V.42bis (here %C)
+DS44	3,0,0,2048,2048, 255,255,6144,6144	V.44 (here %C)
+PCW	2	V.92's call waiting (none is detected here)
+PIG	1	V.92's PCM upstream, an analogue modem's (this one is the digital modem)
+FAA	0	fax adaptive answer (0 only)
+FPR	0	the fax DTE's rate
-SEC	0	Rockwell's MNP 10EC, and its transmit level (1,12)

10.3 Where D-Modem differs

- &K is the compression, as on USRobotics' modems; on Rockwell's and Conexant's it is flow control. &K3 leaves the compression on, &K0 turns it off.
- The S-registers past S16 are D-Modem's own: S29 and S30 are the speaker's (elsewhere the hook flash time and the inactivity timer), S92 the answer's delay (elsewhere a fax transmit level: ATS92=10 makes the modem wait 10 s before its answer tone). The others (S36, S37, S48, S91 . . .) are free: kept as written. S7 and S11 start at 60 and 100.
- The dial string's W pauses 2 s (elsewhere it waits for a second dial tone); ; (back to command mode after dialing), ! (hook flash), @ (quiet answer) and $ (bong tone) are skipped. There are no stored numbers: &Z and ATDS are ERROR.
- ATI's numbers are the modem's own (I0 to I7); ATI8 and up are ERROR.
- The numeric results are 0 to 8 (V0): every CONNECT is 1, its rate in the verbose result only.
- There is no MNP (classes 2 to 5, 10, 10EC): \A is taken and \N1 to \N5 are all V.42. A modem with MNP 10EC connects with V.42 (its "EM" detection patterns are passed over). See docs/link.md, MNP.
- K56flex is not there: +MS=56 runs V.90.

10.4 Not there

ERROR: &V and &Z (no stored profiles or numbers), &T (self tests), \B (a break), X5 and up, -V90 (+MS chooses), +MA, +MSC, +EB, +ESR, +ETBM, +EWIND, +EFRAM, +A8E, +A8T, +PMHF, +GSN, +GOI and #UD; +MS's names of modulations the modem has not got (V32T, V29 . . .).